



Introduction to Software Engineering

Technical Report for "FreeBeats" project:

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Team members:

1. Abdelrahman Hossam 23-101130 (Leader)
2. Arwa Mahmoud 23-10131
3. Rodaina Al-dahshan 23-101193
4. Hana Shoaib 23-101166
5. Mohamed Fathy 23-101167

System vision document:

Problem Description

Most existing music streaming platforms require users to pay for monthly subscriptions or tolerate frequent advertisements that interrupt playback. These limitations reduce user comfort and make the listening experience less enjoyable, especially for people who prefer free and uninterrupted access to music.

In addition, some streaming apps have complex interfaces that are not suitable for users who want a simple and direct experience. Many users simply want to open an app, search for a song, and listen instantly without distractions or unnecessary steps.

Therefore, there is a need for a system that allows users to **listen to songs freely, smoothly, and without interruptions**. The proposed system, **FreeBeats**, addresses this need by offering a clean, ad-free, and user-friendly music streaming experience designed for all music lovers who want simplicity and comfort.

System Capabilities

FreeBeats system will include the following main capabilities:

- Allow users to **search for songs** by name or artist.
- Provide **high-quality audio streaming** with no ads or subscription fees.
- Enable users to **create and manage playlists** (add, remove, rename, save).
- Offer a **Most Played Songs** list that automatically tracks popular tracks.
- Include a **Shuffle Mode** for random playback of songs.
- Maintain a **simple, clean, and intuitive interface** suitable for all users.
- Support **user registration and login** for personalized experiences.
- Ensure **fast performance and stability** even on low-end devices or slow connections.

Business Benefits

Enhance user satisfaction by providing a free, ad-free listening experience.

Attract a large audience of music lovers who prefer simplicity and accessibility.

Improve user engagement with playlist customization and personalized features.

Build a loyal user base, creating opportunities for future system expansion (e.g., analytics or partnerships).

Strengthen the brand image by offering a fast, lightweight, and user-friendly alternative to paid music platforms.

Encourage digital adoption by providing free access to legal, high-quality music streaming

Stakeholder identification:

<i>Stakeholder Category</i>	<i>Examples / Specific Stakeholders</i>	<i>Their Interests in the System</i>
Internal Operational	Developers and Testers - UI/UX Designers - Database Administrators	Ensure the system runs smoothly and without bugs. - Simplify the process of adding new features. - Maintain a user-friendly interface. - Manage data efficiently (songs, users, playlists, etc.)
Internal Executive	Project Manager - Marketing & Business Team - App Owner / Client	Attract a large user base and ensure user retention. - Maintain the system's good reputation and reliability. - Analyze user and music data to support business decisions. - Plan for future scalability (advertisements or premium features)
External Operational	- End Users / Listeners	Access to a simple and intuitive interface. - Enjoy high-quality and fast music streaming. - Create and share personalized playlists. - search on songs or artists
External Executive	Record Labels / Music Production Companies - App Store Platforms (Google Play, Apple Store) - Copyright and Legal Authorities	Protect intellectual property rights. - Ensure legal compliance with music licensing and copyright laws. - Maintain stable and reliable app performance in stores. - Increase visibility and brand reputation of FreeBeats

Functional Requirements

User registration login:

- Users must be able to create an account using email and password
- Users must be able to securely log in and log out
- The system must store user information for personalization.

Song Search and browsing:

- Users must be able to search for songs by name or artist.
- The system must show accurate search results quickly.
- Users must be able to browse recommended or popular songs.

Music playback:

- Users must be able to play, pause, resume and skip tracks.
- The system must provide high-quality audio streaming with no interruptions.
- Playback must continue smoothly even while navigating other pages in the app.

Playlist management:

- Users must be able to create playlists and give them custom names
- Users must be able to add or remove songs from playlists
- Users must be able to view and save multiple personal playlists

Most played & shuffle mode:

- The system must track frequently played songs and display them in a “Most Played” list
- Users must be able to activate shuffle mode for random playback

Recently Played Songs:

- The system must automatically save a history of recently played songs.
- Users must be able to access and replay tracks from the Recently Played list.

Requirements Elicitation Techniques:

Questionnaire

A structured questionnaire was used to collect feedback from potential users and testers.

The questionnaire focused on:

- Frequency of feature usage (search, playback, playlists)
- User experience issues such as delays or playback failures
- Interface usability and overall satisfaction
- Suggestions for feature improvements

Responses helped validate system needs and highlighted priorities, including:

- Fast music playback without interruptions
- Easy to manage playlists
- Improved shuffle functionality
- clear and intuitive interface design

Interviews

Interviews were conducted with both Internal Executive stakeholders (such as project owners and decision-makers) and Internal Operational stakeholders (such as developers and testers).

The goal was to identify:

- The main objective of the FreeBeats system
- Essential system features and priorities
- Expected performance and usability standards
- Challenges that may occur during development and usage

These interviews helped define important features such as:

- Simple and user-friendly interface
- Free and ad-free streaming experience

- Accurate search by song or artist
- Smooth playback and playlist management
- Most played songs feature

Observation and Brainstorming

Observation and brainstorming meetings were conducted among the development team during design and testing phases

Observation and brainstorming focused on :

- Observing user behavior during playback, search and playlist creation
- Discussing design ideas to improve usability and performance
- Identifying possible risks such as search inaccuracy or playback delays

These meetings helped the team to:

- Simplify the overall interface design
- Enhance playback stability and system responsiveness
- Ensure the app remained simple and easy to use across devices

Document Analysis

Document analysis was used to review existing music streaming platforms and related documentation to understand industry standards and user expectations

The analysis focused on:

- Common features provided by other music apps
- Performance standards for audio quality and speed
- Industry standards and design patterns for playlist and search functionality
- Challenges users face with complex interface

Findings from this analysis helped in:

- Making sure the platform provides high quality music streaming
- Designing a simple and user friendly interface
- Identifying the most important features while keeping the system simple

Interview Agendas:

Internal Executive Stakeholder

Setting

Objective of Interview

To discuss the main goals and vision for the FreeBeats system

Date, Time, Location

User Participants

Project Team Participants

Abdalrahman Hossam, Hana Ayman, Mohammed fathy, Arwa Mahmoud, Rodaina Aldahshan

Interview/Discussion

- What is your main goal for the FreeBeats system?
- How do you want the system to benefit users?
- What features do you think are most important?
- How will you measure the system's success?
- How should we handle user feedback and updates?
- What are your future plans for improving FreeBeats?

Follow-Up

Important Decisions or Answers to Questions

- Focus on user-friendly and simple design.
- Keep FreeBeats 100% free and ad-free.
- Success measured by user satisfaction and engagement.
- Add features that make music search easier.

Open Items not resolved with Assignments for solutions

- Approve project budget
- Plan feedback collection method.
- Review launch timeline.
- Choose final design style.
- What challenges or risks do you expect during development?

Internal Operational Stakeholder

Setting

Objective of Interview

To understand the day-to-day operational needs, challenges, and expectations of the FreeBeats system.

Date, Time, Location

User Participants

Project Team Participants

Abdallah Hossam, Hana Ayman, Mohammed fathy, Arwa Mahmoud, Rodaina Aldahshan

Interview/Discussion

- What are your main daily tasks in the system?
- What problems might happen during streaming?
- How should users create and edit playlists?
- How can we keep playback smooth and fast?
- What tools or support do you need for your work?
- What would you change to improve system performance?

Follow-Up

Important Decisions or answers to questions

- Search will work by song name and singer name.
- Focus on smooth and fast playback.
- Playlists will include easy add/remove options.
- “Most Played Songs” will update automatically.

Open Items not resolved with Assignments for solutions

- Confirm tools for maintenance and support.
- Test shuffle mode performance.
- Assign a person to handle database updates.
- Check for streaming bugs.

Questionnaire:

Purpose

This questionnaire aims to collect essential feedback from FreeBeats potential users and team testers to support the requirements analysis process. The data will help identify system strengths, usability issues, and areas that need improvement in the ad-free music streaming platform.

Part I – Answer these questions based on your typical use or work session:

1. How often do you use the search feature to find songs or artists?
2. Have you experienced any interruptions or delays during playback?
3. How easy is it to create and manage playlists?
4. Is the system interface clear and simple to navigate?
5. What improvements would you recommend for a smoother user experience?

Part II – System Evaluation

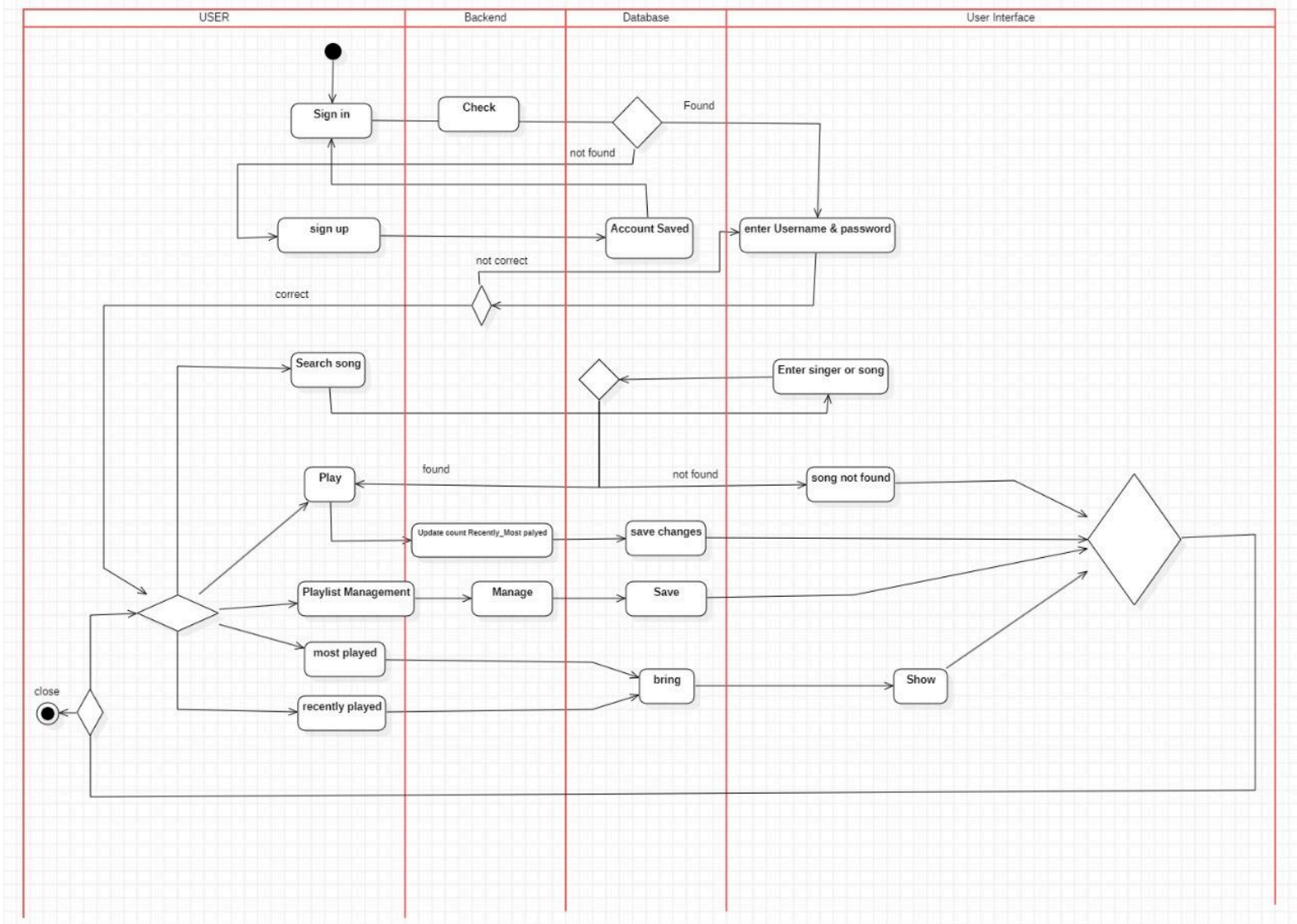
Instructions: Circle the appropriate number on the scale from 1 (Strongly Disagree) to 7 (Strongly Agree) based on how strongly you agree or disagree with each statement.

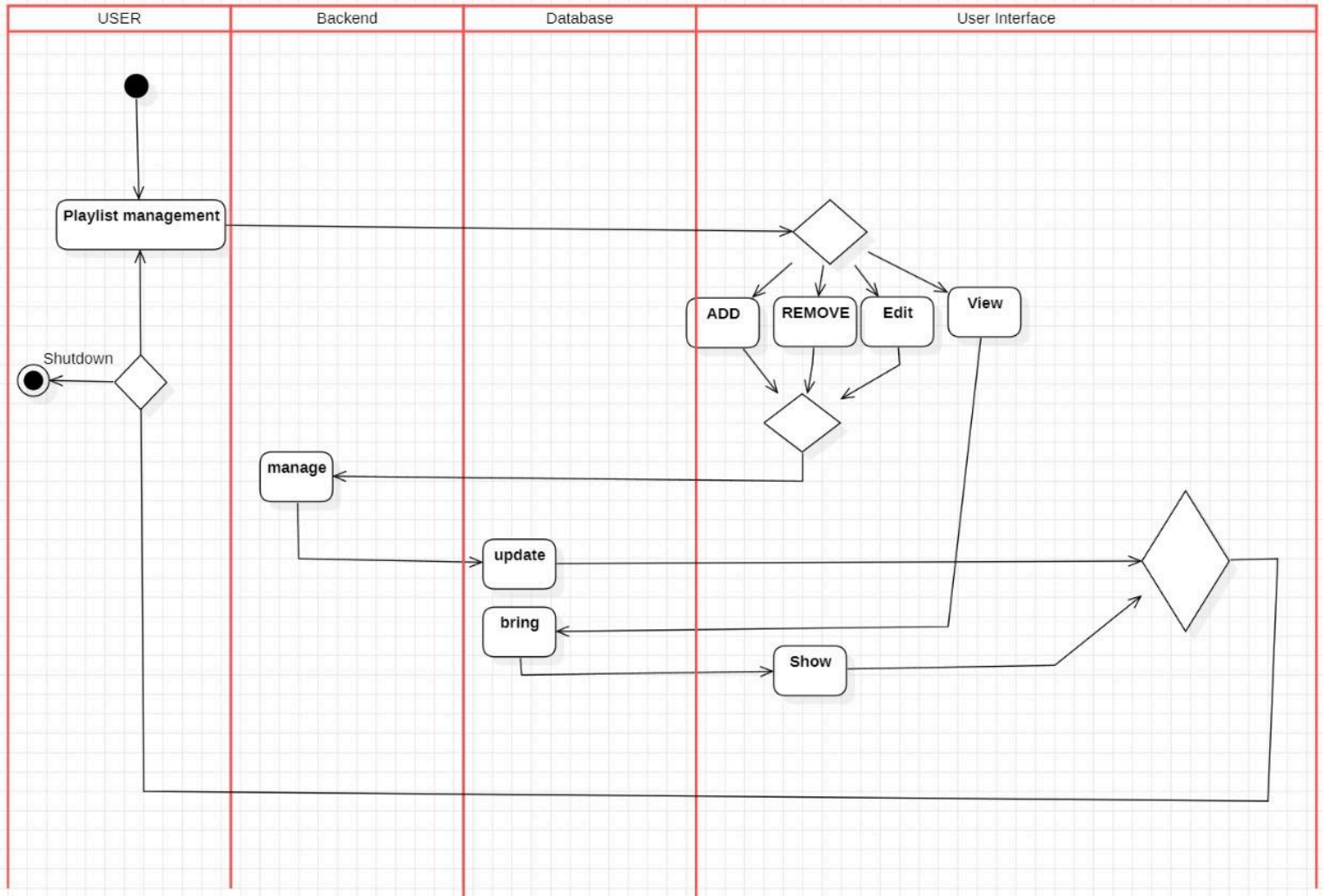
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Part III – Please enter your opinions and comments:

Please describe any issues you encountered and suggest any features or improvements you would like to see in the final version:

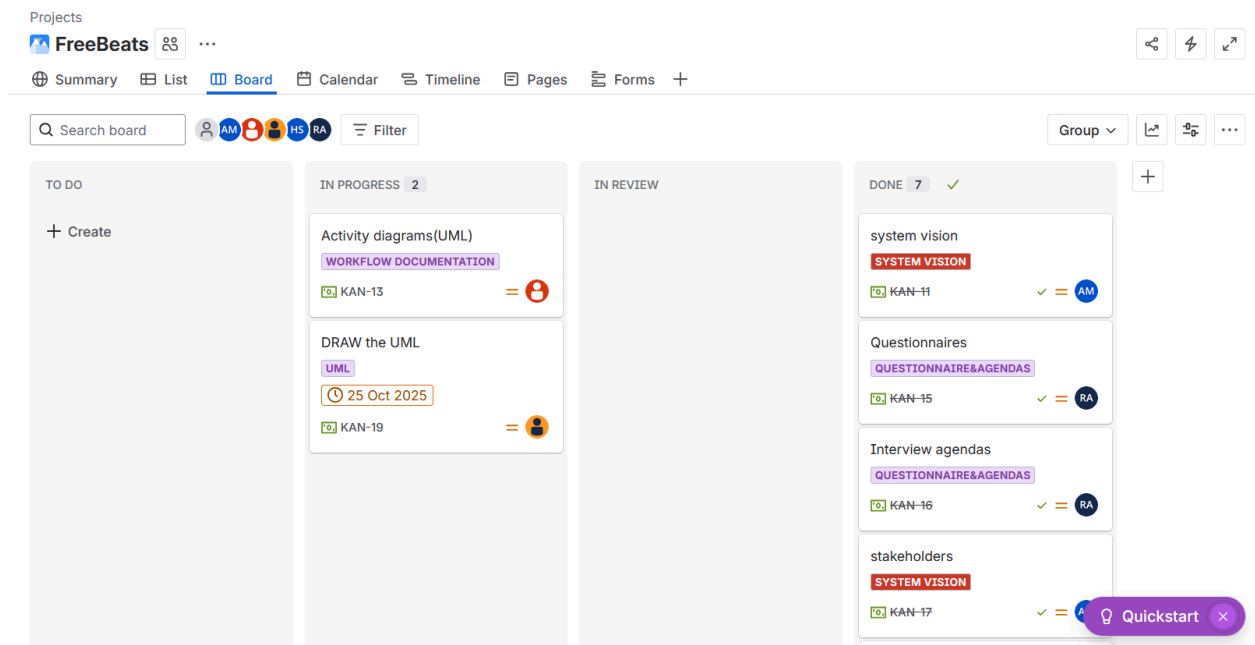
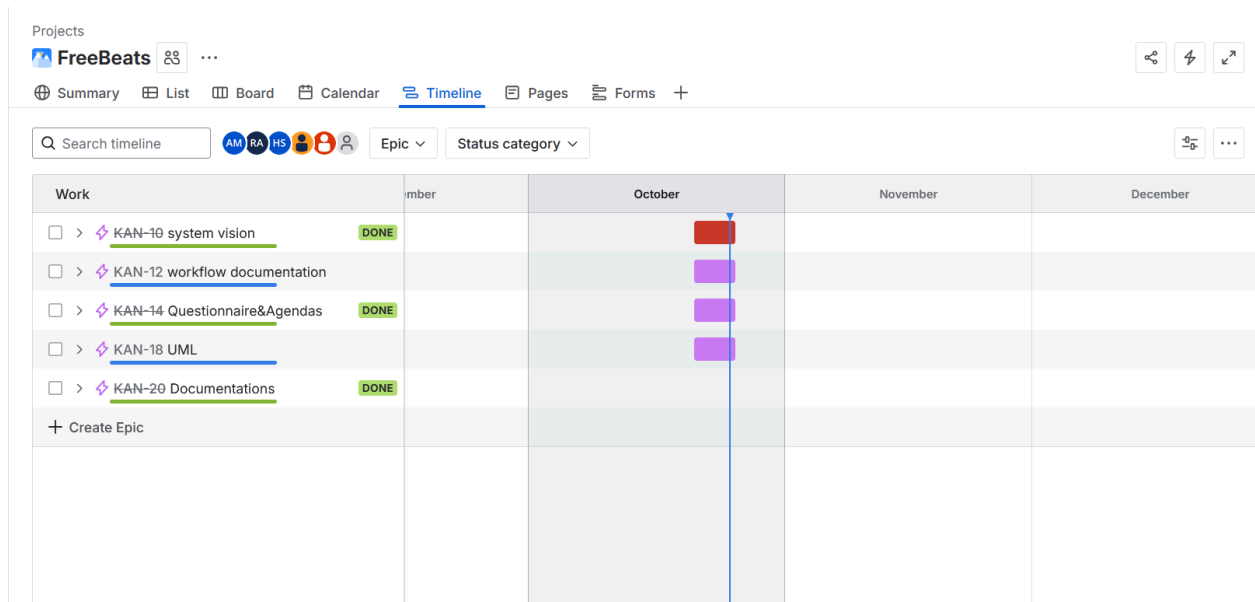
Workflow Documentation:





Jira:

Our team really enjoyed working with **Jira** because it made task management so clear and organized. We could easily see all the tasks in front of us and track the progress together as a team. Every time someone finished their task, they felt a real sense of accomplishment when marking it as “Done” on Jira. Another great thing was that we didn’t need to message each other to say we had completed our work – Jira automatically sent email notifications to everyone. This helped us stay updated and focused without having to constantly remind each other about our tasks.



Github:

<https://github.com/arwa882005/FreeBeats.git>

main
1 Branch
0 Tags
Go to file
Add file
Code

Abdelrhman150 Adding UML 1 ff067ca · now 17 Commits

Functional Requirements.pdf	Add files via upload	2 hours ago
Interview Agenda Executive stakeholder.pdf	Rename Interview Agendas-executive stakeholder.pdf to Int...	1 hour ago
Interview Agenda operational stakeholder.pdf	Rename FreeBeats_Interview_Agendas operational stakehold...	1 hour ago
Questionnaire.pdf	Adding Questionnaire	3 hours ago
Requirements Elicitation Techniques.pdf	Add files via upload	2 hours ago
Stakeholder identification.pdf	Add files via upload	3 hours ago
System vision document.pdf	Add files via upload	3 hours ago
UML1mdj.mdj	Adding UML 1	now
UML2mdj.mdj	Adding UML 2 -playlist management-	9 minutes ago

README



Add a README

Help people interested in this repository understand your project by adding a README.

About

FreeBeats : is a complete music streaming system that lets users listen to songs for free with no ads, no subscriptions, and no interruptions.

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Abdelrhman150 committed 1 hour ago 7f26f3b

Rename Interview Agendas-executive stakeholder.pdf to Interview Agenda Executive stakeholder.pdf

Rodaina-Aldahshan authored 1 hour ago Verified a3c1cfc

Rename FreeBeats_Interview_Agendas operational stakeholder.pdf to Interview Agenda operational stakeholder.pdf

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hana-shoail authored 3 hours ago Verified 6896742

Add files via upload

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Adding Questionnaire

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